

2024 Calculus II Additional Exercise (11)

1. Exercise 1, 2, 3, 6(first), 8, 13 in page 46 from Calculus Supplement Homework.
2. Exercise 17, (6), (7), (14)($\iiint$ is revised as $\iint$), (16) in page 47 from Calculus Supplement Homework.
3. Exercise 40 in page 50 from Calculus Supplement Homework.
4. Find

$$J = \int_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}} \int_{|x|}^{\sqrt{1-x^2}} \sqrt{1-y^2} \, dy \, dx.$$

5. Evaluate $\iiint_D \frac{x^2}{a^2} \, dV$, where $D : \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \leq 1$ is an ellipsoid.

Q1 P46 Ex 1, 2, 3, 6, 8, 13

1. Let $D = \{(x, y) \mid x^2 + y^2 \leq 4\}$, then $\iint_D \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} \right) dA = (\quad)$.
2. Let $D = \{(x, y) \mid x^2 + y^2 \leq x + y\}$, then in polar coordinate $\iint_D f(x, y) dx dy = (\quad)$.
3. Let Ω be bounded by $z = \sqrt{x^2 + y^2}$ and $x^2 + y^2 + z^2 = 2z$. In spherical coordinates, the triple integral $\iiint_{\Omega} (x^2 + y^2 + z^2) dV = (\quad)$.

Sol to Ex1:

$$\text{原式} = \int_0^{2\pi} \int_0^2 r^2 \left(\frac{\cos^2 \theta}{a^2} + \frac{\sin^2 \theta}{b^2} \right) \cdot r dr d\theta$$

$$= \int_0^{2\pi} 4 \left(\frac{1 + \cos 2\theta}{2a^2} + \frac{1 - \cos 2\theta}{2b^2} \right) d\theta$$

$$= 4\pi \left(\frac{1}{a^2} + \frac{1}{b^2} \right) + 4 \left(\frac{1}{2a^2} - \frac{1}{2b^2} \right) \frac{1}{2} \sin 2\theta \Big|_0^{2\pi}$$

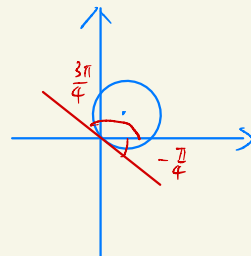
$$= \frac{4\pi}{a^2} + \frac{4\pi}{b^2}$$

□

Sol to Ex2:

$$D = \{(x, y) \mid x^2 + y^2 \leq x + y\} = \{(r, \theta) \mid 0 \leq r \leq \cos \theta + \sin \theta, -\frac{\pi}{4} \leq \theta \leq \frac{3\pi}{4}\}$$

$$\therefore \iint_D f(x, y) dx dy = \int_{-\frac{\pi}{4}}^{\frac{3\pi}{4}} \int_0^{\cos \theta + \sin \theta} r f(r \cos \theta, r \sin \theta) dr$$



□

Sol to Ex3:

$$\Omega = \{(x, y, z) \mid \sqrt{x^2 + y^2} \leq z, z \geq \frac{x^2 + y^2 + z^2}{2}\}$$

$$= \{(\rho, \varphi, \theta) \mid \rho \sin \varphi \leq \rho \cos \varphi \leq \frac{\rho^2}{2}\}$$

$$= \{(\rho, \varphi, \theta) \mid 2 \cos \varphi \geq \rho \geq 0, 0 \leq \varphi \leq \frac{\pi}{4}\}$$

$$\therefore \text{原式} = \iiint_{\Omega} x^2 + y^2 + z^2 dV$$

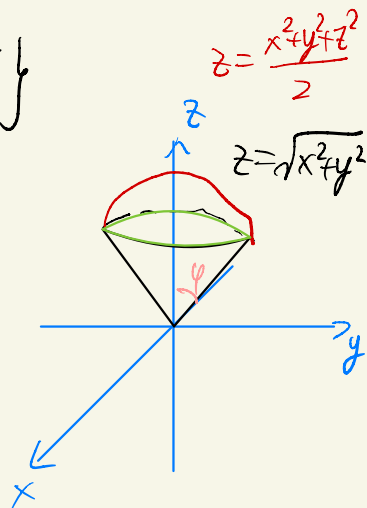
$$= \int_0^{2\pi} \int_0^{\frac{\pi}{4}} \int_0^{2 \cos \varphi} \rho^2 \cdot \rho^2 \sin \varphi d\rho d\varphi d\theta$$

$$= \int_0^{2\pi} \int_0^{\frac{\pi}{4}} \frac{2^5 \cos^5 \varphi \cdot \sin \varphi}{5} d\varphi d\theta$$

$$= \frac{32}{5} \int_0^{2\pi} \left(-\frac{\cos^6 \varphi}{6} \Big|_0^{\frac{\pi}{4}} \right) d\theta$$

$$= \frac{32}{5} \cdot 2\pi \cdot \left(-\frac{1}{6} \right) \left(\frac{1}{2^3} - 1 \right)$$

$$= \frac{28}{15} \pi$$



$$\begin{cases} z = \sqrt{x^2 + y^2} \\ 2z = x^2 + y^2 + z^2 \end{cases}$$

$$\Rightarrow \begin{cases} z=0 \text{ (原点)} \\ z=1 \text{ (图中圆圈)} \end{cases}$$

□

6. Interchange the order of the integral:

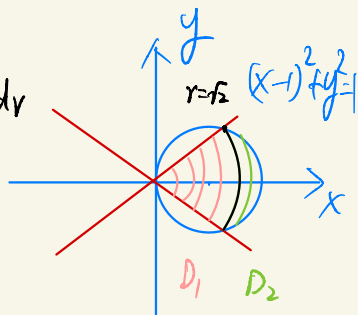
$$\int_{-\pi/4}^{\pi/4} \int_0^{2\cos\theta} f(r, \theta) r \, dr \, d\theta = (\quad);$$

Sol.: $D = \{ (r, \theta) \mid 0 \leq r \leq 2\cos\theta, -\frac{\pi}{4} \leq \theta \leq \frac{\pi}{4} \}$

$$= \{ (x, y) \mid (x-1)^2 + y^2 \leq 1, -x \leq y \leq x \}$$

拆分成 D_1 和 D_2

$$\text{原式} = \int_0^{\sqrt{2}} \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} f(r, \theta) r \, d\theta \, dr + \int_{\sqrt{2}}^2 \int_{-\cos^{-1}(\frac{r}{2})}^{\cos^{-1}(\frac{r}{2})} f(r, \theta) r \, d\theta \, dr$$



8. Let $f(x, y)$ be continuous and positive. Then $\lim_{n \rightarrow \infty} \int_0^\pi \int_0^\pi (f(x, y))^{1/n} dx dy = (\quad)$.

Sol: $\exists m, M \in \mathbb{R}$, s.t. $\forall (x, y) \in [0, \pi] \times [0, \pi] \quad m \leq f(x, y) \leq M$

$$\therefore \int_0^\pi \int_0^\pi m^{\frac{1}{n}} dx dy \leq \int_0^\pi \int_0^\pi (f(x, y))^{\frac{1}{n}} dx dy \leq \int_0^\pi \int_0^\pi M^{\frac{1}{n}} dx dy$$

$$\parallel \\ m^{\frac{1}{n}} \pi^2$$

$$\downarrow n \rightarrow \infty \\ \pi^2$$

$$\parallel \\ M^{\frac{1}{n}} \pi^2$$

$$\downarrow n \rightarrow \infty \\ \pi^2$$

$$\therefore \text{原式} = \pi^2$$

□

13. Write the triple integral in order of $dx dz dy$: $\int_0^1 dx \int_0^{1-x} dy \int_0^{x+y} f(x, y, z) dz = (\quad)$.

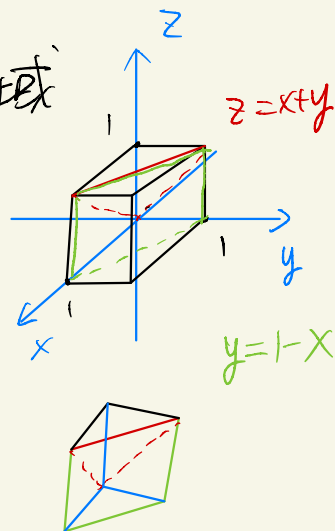
Sol:

$$\{z = x + y\} \quad \{y = 1 - x\}$$

积分区域: 红、绿和坐标平面围成的区域

$$\therefore \text{原式} = \int_0^1 \int_0^{1-y} \int_0^{1-y} f(x, y, z) dx dz dy$$

$$= \int_0^1 \int_y^1 \int_0^{z-y} f(x, y, z) dx dz dy$$



□

Q2 P47 Ex 17 (6), (7), (14), (16)

$$(5) \iiint_{\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \leq 1} e^{\sqrt{\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2}}} dV;$$

$$(6) \iint_{x^2+y^2 \leq 1, x \geq 0} \frac{1+xy}{1+x^2+y^2} dA;$$

$$(7) \iiint_{\substack{x^2+y^2+z^2 \geq z \\ x^2+y^2+z^2 \leq 2z}} z dV;$$

$$(8) \iiint_{x^2+y^2+z^2 \leq R^2} \sqrt{x^2+y^2+z^2} dV;$$

$$(9) \star \iint_{\substack{x+y \leq 1 \\ x, y \geq 0}} e^{\frac{y}{x+y}} dx dy;$$

$$(10) \iint_{\substack{0 \leq x \leq 1 \\ 0 \leq y \leq 1}} |x^2 + y^2 - 1| dA;$$

$$(11) \int_0^1 \int_{3y}^1 e^{x^2} dx dy;$$

$$(12) \iint_{\substack{x^2+y^2 \leq 9 \\ 0 \leq y \leq \sqrt{3}x}} (x^2 + y^2)^{3/2} dA;$$

$$(13) \iint_{|x|+|y| \leq 1} (|x| + ye^{-x^2}) dA;$$

$$(14) \iiint_{x^2+y^2 \leq 4} |x^2 + y^2 - 2y| dA;$$

$$(15) \iint_{x^2+y^2 \leq x+y} (x+y) dA;$$

$$(16) \iint_{|x|+|y| \leq 1} e^{x+y} dA;$$

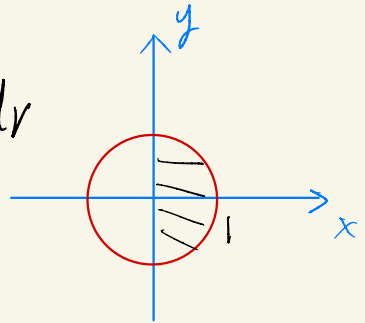
Sol to (6)

$$\text{原式} = \int_0^1 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1+r^2 \sin \theta \cos \theta}{1+r^2} r d\theta dr$$

$$= \int_0^1 \frac{\pi r}{1+r^2} dr$$

$$= \frac{\pi}{2} \ln(1+r^2) \Big|_0^1$$

$$= \frac{\pi}{2} \ln 2$$

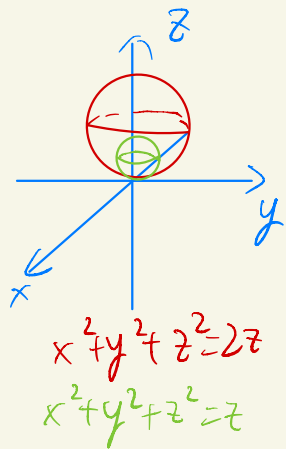


Sol to (7)

$$\Omega = \{(x, y, z) \mid z \leq x^2 + y^2 + z^2 \leq 2z\}$$

$$= \{(x, y, z) \mid r \cos \varphi \leq r^2 \leq 2r \cos \varphi\}$$

$$= \{(x, y, z) \mid \cos \varphi \leq r \leq 2 \cos \varphi, 0 \leq \varphi \leq \frac{\pi}{2}\}$$



$$\text{原式} = \iiint_{\Omega} z \, dv$$

$$= \int_0^{2\pi} \int_0^{\frac{\pi}{2}} \int_{\cos \varphi}^{2 \cos \varphi} r \cos \varphi \cdot r^2 \sin \varphi \, dr \, d\varphi \, d\theta$$

$$= \int_0^{2\pi} \int_0^{\frac{\pi}{2}} \cos \varphi \sin \varphi \cdot \frac{1}{4} [(2 \cos \varphi)^4 - \cos^4 \varphi] \, d\varphi \, d\theta$$

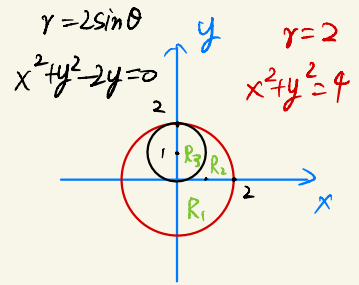
$$= \frac{15}{4} \int_0^{2\pi} \left(-\frac{\cos^6 \varphi}{6} \right) \Big|_0^{\frac{\pi}{2}} d\theta$$

$$= \frac{15}{4} \cdot 2\pi \cdot \left(-\frac{1}{6} \right) (0 - 1)$$

$$= \frac{5}{4} \pi$$

□

Sol to (14):



$$\begin{aligned} f(x, y) &= x^2 + y^2 - 2y \\ &= x^2 + (y-1)^2 - 1 \end{aligned}$$

$$\begin{aligned} \iint_{R_1} |f(x, y)| dA &= \int_{-\frac{\pi}{2}}^0 \int_0^2 (r^2 - 2r \sin \theta) r dr d\theta = \int_{-\frac{\pi}{2}}^0 \left(\frac{r^4}{4} - \frac{2r^3}{3} \sin \theta \right) \Big|_0^2 d\theta \\ &= \int_{-\frac{\pi}{2}}^0 4 - \frac{16}{3} \sin \theta d\theta \\ &= 2\pi + \frac{16}{3} \cos \theta \Big|_{-\frac{\pi}{2}}^0 \\ &= 2\pi + \frac{16}{3} \end{aligned}$$

$$\begin{aligned} \iint_{R_2} |f(x, y)| dA &= \int_{\frac{\pi}{2}}^{\pi} \int_{2\sin \theta}^2 (r^2 - 2r \sin \theta) r dr d\theta = \int_{\frac{\pi}{2}}^{\pi} \left(\frac{r^4}{4} - \frac{2r^3}{3} \sin \theta \right) \Big|_{2\sin \theta}^2 d\theta \\ &= \int_{\frac{\pi}{2}}^{\pi} 4 - 4\sin^4 \theta - \frac{16}{3} \sin \theta + \frac{16}{3} \sin^4 \theta d\theta \\ &= 2\pi + \int_{\frac{\pi}{2}}^{\pi} \frac{4}{3} \cdot \left(\frac{1 - \cos 2\theta}{2} \right)^2 - \frac{16}{3} \sin \theta d\theta \\ &= 2\pi + \frac{1}{3} \int_{\frac{\pi}{2}}^{\pi} 1 - 2\cos 2\theta + \frac{1 + \cos 4\theta}{2} - 16\sin \theta d\theta \\ &= 2\pi + \frac{\pi}{4} + \frac{1}{3} \left(-\sin 2\theta + \frac{\sin 4\theta}{8} + 16\cos \theta \right) \Big|_{\frac{\pi}{2}}^{\pi} \\ &= \frac{9}{4}\pi - \frac{16}{3} \end{aligned}$$

$$\begin{aligned}
\iint_{R^1} |f(x,y)| dA &= \int_0^{\frac{\pi}{2}} \int_0^{2\sin\theta} (2r\sin\theta - r^2) r dr d\theta = \int_0^{\frac{\pi}{2}} \left(\frac{2r^3}{3} \sin\theta - \frac{r^4}{4} \right) \Big|_0^{2\sin\theta} d\theta \\
&= \int_0^{\frac{\pi}{2}} \frac{16}{3} \sin^4\theta - 4\sin^4\theta d\theta \\
&= \frac{4}{3} \int_0^{\frac{\pi}{2}} \frac{(1-\cos 2\theta)^2}{4} d\theta \\
&= \frac{1}{3} \int_0^{\frac{\pi}{2}} 1 - 2\cos 2\theta + \frac{1+\cos 4\theta}{2} d\theta \\
&= \frac{\pi}{4} + \frac{1}{3} \left(-2\sin 2\theta + \frac{\sin 4\theta}{8} \right) \Big|_0^{\frac{\pi}{2}} \\
&= \frac{\pi}{4}
\end{aligned}$$

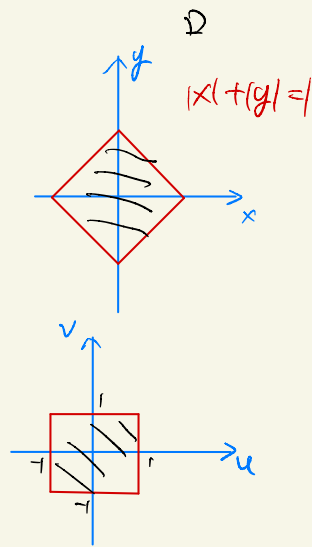
$$\begin{aligned}
\therefore \text{原式} &= 2 \left(\iint_{R_1} + \iint_{R_2} + \iint_{R_3} \right) |f(x,y)| dA \\
&= 2 \left(2\pi + \frac{16}{3} + \frac{9}{4}\pi - \frac{16}{3} + \frac{\pi}{4} \right) \\
&= 9\pi
\end{aligned}$$

Sol to (1b):

$$\text{换元} \begin{cases} u = x+y \\ v = -x+y \end{cases} \Rightarrow \begin{cases} x = \frac{u-v}{2} \\ y = \frac{u+v}{2} \end{cases}$$

$$\left| \frac{\partial(x,y)}{\partial(u,v)} \right| = \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial x}{\partial v} \frac{\partial y}{\partial u} = \frac{1}{4} - \left(-\frac{1}{4} \right) = \frac{1}{2}$$

$$\text{New boundary conditions: } \begin{aligned} u &= 1 \\ u &= -1 \\ v &= 1 \\ v &= -1 \end{aligned}$$



$$\therefore \text{原式} = \int_{-1}^1 \int_{-1}^1 e^{\frac{u}{2}} du dv = e - \frac{1}{e}$$

Q3 P50 Ex 40

40. ★ Given $F(t) = \int_1^t \int_y^t f(x) dx dy$ where f is a single variable continuous function. Compute $F'(2)$.

Sol: let $g(t, y) = \int_y^t f(x) dx$, then $F(t) = \int_1^t g(t, y) dy$

$$\begin{aligned}\therefore F'(t) &= g(t, t) + \int_1^t \frac{\partial g(t, y)}{\partial t} dy \\ &= 0 + \int_1^t f(t) dy \\ &= f(t)(t-1)\end{aligned}$$

$$F'(2) = f(2)$$

□

Q4

4. Find

$$J = \int_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}} \int_{|x|}^{\sqrt{1-x^2}} \sqrt{1-y^2} dy dx.$$

sol: (法一: 先对 y 积分, 还挺算!)

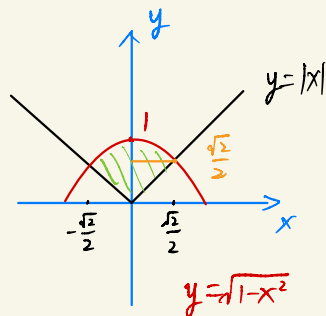
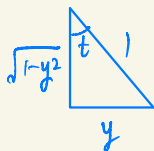
compute: $\int \sqrt{1-y^2} dy$

$$\stackrel{y=\sin t}{=} \int \cos^2 t dt$$

$$= \int \frac{1+\cos 2t}{2} dt$$

$$= \frac{t}{2} + \frac{\sin 2t}{4} + C$$

$$= \frac{\arcsin y}{2} + \frac{y\sqrt{1-y^2}}{2} + C$$



$$\therefore J = 2 \int_0^{\frac{\sqrt{2}}{2}} \int_x^{\sqrt{1-x^2}} \sqrt{1-y^2} dy dx$$

$$= \int_0^{\frac{\sqrt{2}}{2}} \arcsin \sqrt{1-x^2} - \arcsin x + \sqrt{1-x^2} x - x\sqrt{1-x^2} dx$$

$$= x(\arcsin \sqrt{1-x^2} - \arcsin x) \Big|_0^{\frac{\sqrt{2}}{2}} - \int_0^{\frac{\sqrt{2}}{2}} x \cdot \left(\frac{-2x}{\sqrt{1-x^2} \cdot 2\sqrt{1-x^2}} - \frac{1}{\sqrt{1-x^2}} \right) dx$$

$$= \int_0^{\frac{\sqrt{2}}{2}} \frac{2x}{\sqrt{1-x^2}} dx$$

$$\stackrel{u=x^2}{=} \int_0^1 \frac{1}{\sqrt{1-u}} du$$

$$= -2\sqrt{1-u} \Big|_0^1$$

$$= 2 - \sqrt{2}$$

(法二: 先对 x 积分, 相对容易算!)

$$J = 2 \left[\int_0^{\frac{\sqrt{2}}{2}} \int_0^y \sqrt{1-y^2} dx dy + \int_{\frac{\sqrt{2}}{2}}^1 \int_0^{\sqrt{1-y^2}} \sqrt{1-y^2} dx dy \right]$$

$$= 2 \left[\int_0^{\frac{\sqrt{2}}{2}} y \sqrt{1-y^2} dy + \int_{\frac{\sqrt{2}}{2}}^1 1-y^2 dy \right]$$

$$= \int_0^{\frac{\sqrt{2}}{2}} \sqrt{1-y^2} dy^2 + 2 \left(y - \frac{y^3}{3} \right) \Big|_{\frac{\sqrt{2}}{2}}^1$$

$$\stackrel{u=y^2}{=} = \frac{-2}{3} (1-u)^{\frac{3}{2}} \Big|_0^{\frac{1}{2}} + 2 \left(1 - \frac{1}{3} - \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{12} \right)$$

$$= -\frac{2}{3} \left(\frac{\sqrt{2}}{4} - 1 \right) + 2 \left(\frac{2}{3} - \frac{5\sqrt{2}}{12} \right)$$

$$= 2 - \sqrt{2}$$

(法三 极坐标在后面!)

□

Q4.

4. Find

$$J = \int_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}} \int_{|x|}^{\sqrt{1-x^2}} \sqrt{1-y^2} dy dx.$$

Sol: (法三: 极坐标, 开根号时小心!)

$$J = \int_{\frac{\pi}{4}}^{\frac{3}{4}\pi} \int_0^1 r \sqrt{1-r^2 \sin^2 \theta} dr d\theta$$

$$= \frac{1}{2} \int_{\frac{\pi}{4}}^{\frac{3}{4}\pi} \int_0^1 \sqrt{1-r^2 \sin^2 \theta} dr^2 d\theta$$

$$= \frac{1}{2} \int_{\frac{\pi}{4}}^{\frac{3}{4}\pi} \left(\frac{-2(1-r^2 \sin^2 \theta)^{\frac{3}{2}}}{3 \sin^2 \theta} \Big|_0^1 \right) d\theta$$

$$= \frac{1}{3} \int_{\frac{\pi}{4}}^{\frac{3}{4}\pi} \frac{-(\cos^2 \theta)^{\frac{3}{2}}}{\sin^2 \theta} + \frac{1}{\sin^2 \theta} d\theta$$

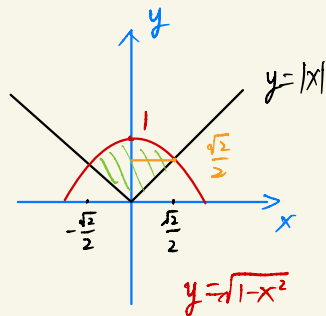
$$= \frac{1}{3} \cdot (-\cot \theta) \Big|_{\frac{\pi}{4}}^{\frac{3}{4}\pi} + \frac{1}{3} \left(\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{-\cos^3 \theta}{\sin^2 \theta} d\theta + \int_{\frac{\pi}{2}}^{\frac{3}{4}\pi} \frac{\cos^3 \theta}{\sin^2 \theta} d\theta \right)$$

$$= \frac{2}{3} + \frac{1}{3} \left[\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{\sin^2 \theta - 1}{\sin^2 \theta} d\sin \theta + \int_{\frac{\pi}{2}}^{\frac{3}{4}\pi} \frac{1 - \sin^2 \theta}{\sin^2 \theta} d\sin \theta \right]$$

$$= \frac{2}{3} + \frac{1}{3} \left[\left(\sin \theta + \frac{1}{\sin \theta} \right) \Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}} + \left(-\sin \theta - \frac{1}{\sin \theta} \right) \Big|_{\frac{\pi}{2}}^{\frac{3}{4}\pi} \right]$$

$$= \frac{2}{3} + \frac{1}{3} \left[\left(2 - \frac{\sqrt{2}}{2} - \sqrt{2} + \left(-\frac{\sqrt{2}}{2} - \sqrt{2} + 2 \right) \right) \right]$$

$$= 2 - \sqrt{2}$$



Q5

5. Evaluate $\iiint_D \frac{x^2}{a^2} dV$, where $D: \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \leq 1$ is an ellipsoid.

Sol. ~~转换~~:

$$\begin{cases} x = ar \sin \varphi \cos \theta \\ y = br \sin \varphi \sin \theta \\ z = cr \cos \varphi \end{cases}$$

$$\left| \frac{\partial(x, y, z)}{\partial(r, \varphi, \theta)} \right| = \begin{vmatrix} a \sin \varphi \cos \theta & ar \cos \varphi \cos \theta & -ar \sin \varphi \sin \theta \\ b \sin \varphi \sin \theta & br \cos \varphi \sin \theta & br \sin \varphi \cos \theta \\ c \cos \varphi & -cr \sin \varphi & 0 \end{vmatrix}$$

$$= abc r^2 \begin{vmatrix} \sin \varphi \cos \theta & \cos \varphi \cos \theta & -\sin \varphi \sin \theta \\ \sin \varphi \sin \theta & \cos \varphi \sin \theta & \sin \varphi \cos \theta \\ \cos \varphi & -\sin \varphi & 0 \end{vmatrix}$$

$$= abc r^2 [\cos \varphi \cdot \sin \varphi \cos \varphi + \sin \varphi \sin \varphi \sin \varphi]$$

$$= abc r^2 \sin \varphi$$

$$\therefore \iiint_D \frac{x^2}{a^2} dV = \int_0^{2\pi} \int_0^\pi \int_0^1 abc r^2 \sin^2 \varphi \cos^2 \theta \cdot r^2 \sin \varphi dr d\varphi d\theta$$

$$= abc \left(\frac{r^5}{5} \Big|_0^1 \right) \cdot \int_0^\pi (\cos^2 \varphi - 1) d\varphi \cdot \int_0^{2\pi} \frac{1 + \cos 2\theta}{2} d\theta$$

$$= \frac{abc}{5} \cdot \left(\frac{\cos^3 \varphi}{3} - \cos \varphi \right) \Big|_0^\pi \cdot \pi$$

$$= \frac{abc\pi}{5} \cdot 2 \left(1 - \frac{1}{3} \right)$$

$$= \frac{4abc\pi}{15}$$

